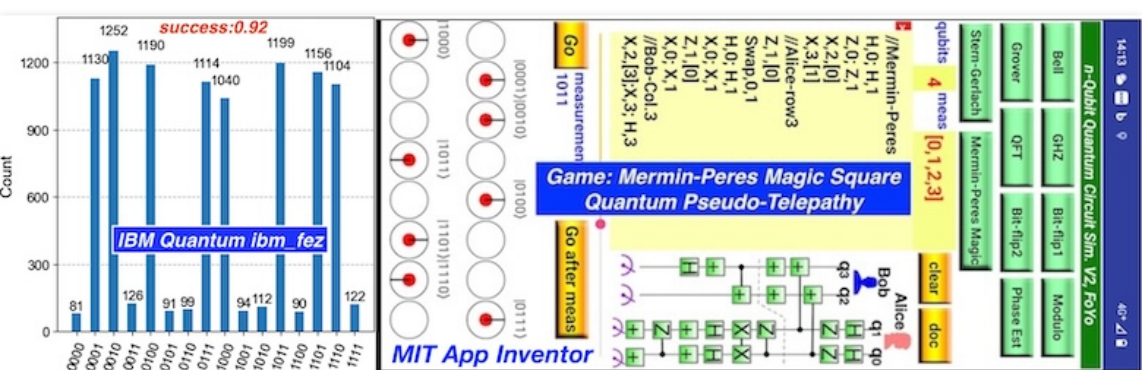




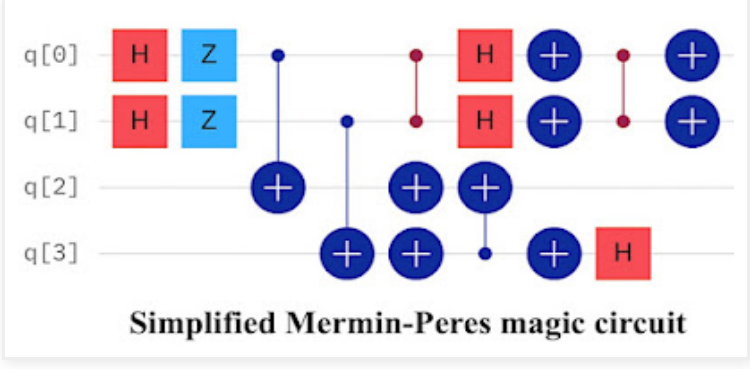
I am a professor emeritus of CS at Kanagawa Institute of Technology, Japan. Originally my specialty was parallel and distributed systems. My current interests include machine learning, natural language processing, creating mobile apps with MIT App Inventor, and quantum computing. In the web version of this blog, clicking the icon on the right (a plastic sphere) will take you to the "List of Quantum Computing Articles". - Fujio Yamamoto (for e-mail, add "@ieee.org" after "yamamotof")

2025年11月10日月曜日

The Evolution of IBM Quantum Hardware

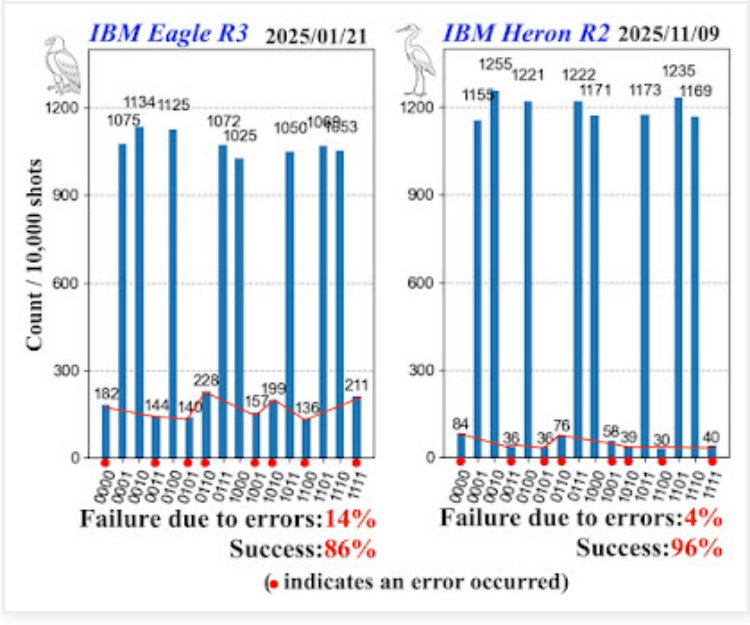
IBM Quantum has taken another big step forward. In addition to the familiar Eagle R3 processor, the company has now made its latest Heron R2 processor available to Open Plan (free-tier) users. I tested the following quantum circuit on Heron R2, a circuit where quantum entanglement plays a key role.

IBM Quantumは、これまでのEagle R3プロセッサに加えて、最新機Heron R2を、Open Plan（無償）ユーザに提供を開始した。私は、以下に示した量子回路（量子もつれが重要な役割を果たしている）をHeron R2で実行してみた。



The result was astonishing. As shown in the figure, the error rate dropped to less than one-third of what I observed ten months ago when running the same circuit on Eagle R3. Thanks to this improvement, the success rate jumped from 86% to an impressive 96%. For users to feel this level of hardware progress firsthand is incredibly meaningful. Thank you, IBM Quantum!

驚くべき結果であった！下図に示す通り、10ヶ月前にEagle R3プロセッサで実行した場合よりも、エラー率が1/3以下に低減した！これにより、この量子回路での正答率は86%から96%に大幅に向上した。ユーザが、ハードウェアのこのような進歩を実感できたことの意味は非常に大きい。ありがとう、IBM Quantum!



By the way, when you look at the processor logos for Eagle and Heron, most people probably feel that the Eagle looks stronger and more powerful. But in reality, Herons are remarkably resilient and graceful. That “grace” might very well symbolize the reduction in errors. IBM really nailed the naming on this one!

ところで、プロセッサのロゴ、ワシ（Eagle）とサギ（Heron）を比較すると、ワシの方が強そうなイメージを持つ人が多いだろう。しかし、実はサギもしなやかで強いのである。「しなやかさ」は、エラー低減を意味するのかもしれない。IBMは良いネーミングをしたものですね！

urbana 時刻: 7:28

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